4 Experiments

This section details the datasets, evaluation metrics, models, and implementation specifics used to validate our proposed two-stage methodology for screenplay generation.

4.1 Experimental Setting

Datasets. Our experiments utilize two primary datasets: a large-scale dataset for fine-tuning and a custom high-quality dataset for evaluation.[2]

- ***Fine-Tuning Dataset***: We constructed a comprehensive fine-tuning dataset comprising $50,000$ samples covering diverse genres.
- ***Evaluation Dataset***: We built a high-quality test set tailored for screenplay evaluation, comprising 32 distinct scenes sourced from four different television series, carefully curated by human experts.

Evaluation Metrics. Given the creative and subjective nature of screenplay writing, our evaluation relies on expert human judgment, supplemented by comparative and diagnostic metrics.[3]

The primary evaluation metric is a holistic quality score assigned by professional Chinese television drama screenwriters. We involved over 20 experienced screenwriters in the evaluation, with each screenplay independently scored by all of them. The final score for each screenplay is the average of all screenwriters' ratings. The scoring rubric, illustrated in Figure 3, employs a 12-point scale organized into six quality tiers.

In addition, we designed several auxiliary metrics including Variance, Error Counts, Ratio to Human, and Win Rate to enable a more comprehensive analysis of model performance.

Models for Comparison. The experimental design encompasses a diverse set of models for comprehensive evaluation, including both fine-tuned models and proprietary state-of-the-art APIs.[4]

- ***Fine-Tuned Models***: Several models of varying scales and specializations are compared, including `Generic-LLM` (`Qwen-14B-Chat` and `QwQ-32B`), `Qwen-72B-Chat`, and `Qwen-72B-CPT`.
- ***API Models***: The best-performing fine-tuned model is evaluated against `Claude-Sonnet-4` and `Gemini-2.5-Pro`.

Implementation Details. To ensure a fair and rigorous comparison, all fine-tuned models were trained on the same dataset using identical hyperparameters.[5]

4.2 Main Results

Table 1 presents the evaluation results. Compared to the base model `Qwen-72B-Chat` and state-of-the-art LLMs `Gemini-2.5-Pro` and `Claude-Sonnet-4`, screenplays generated by our DSR framework achieve a higher win rate, lower variance, and attain the highest average score of 8.06. Although this remains below the human-written reference score of 9.75, reaching approximately 83% of professional quality, the generated scripts are sufficiently high-quality to serve as viable first drafts for further refinement by professional writers.

Furthermore, Figure 4 illustrates the frequency of different error types across various models. Our DSR framework substantially reduces error rates compared to `Gemini-2.5-Pro`, particularly in character development and narrative pacing. This indicates that DSR strengthens the model's capacity to construct consistent and psychologically coherent characters, while also enabling the generation of scripts with better narrative pacing. Additionally, the DSR approach demonstrates superior controllability in text generation, exhibiting minimal deviation from the input outline, which suggests stronger alignment between the generated content and the intended narrative structure.

These results show that our method enhances LLMs' performance in screenplay generation, particularly in managing narrative structure, character development, and dramatic expression. This advancement represents a meaningful step toward practical AI-assisted screenwriting, bringing automated text generation closer to professional storytelling standards.

4.3 Ablation Study

To validate the design choices of our DSR framework, we conduct comprehensive ablation studies.[6] The key findings are: (1) Model scale is

[2] See Appendix B.1 for detailed dataset statistics and composition.

[3] See Appendix B.2 for complete metric definitions and scoring rubric details.

[4] See Appendix B.3 for complete model specifications.

[5] See Appendix B.4 for complete hyperparameters and hardware specifications.

[6] See Appendix B.5 for complete ablation results and detailed analysis.

Table 1: Evaluation results of different models on screenplay generation. The *Expert Score* is averaged across ratings from over 20 professional screenwriters. Red superscripts with upward arrows indicate improvement over the baseline (`Qwen-72B-Chat`). Bold numbers indicate the best performance among models in each metric.

Model	**Method**	**Expert Score**	**Variance**	**Ratio to Human** (%)	**Win Rate** (%)
`Qwen-72B-Chat`	Prompt	3.43	0.19	35.18	0.0
`Claude-Sonnet-4`	Prompt	$6.69^{\uparrow 3.26}$	0.41	68.61	12.5
`Gemini-2.5-Pro`	Prompt	$6.95^{\uparrow 3.52}$	0.34	71.28	12.5
`Qwen-72B-CPT`	DSR (Ours)	$\mathbf{8.06}^{\uparrow 4.63}$	**0.14**	**82.67**	**75.0**
Human	-	9.75	0.08	-	-

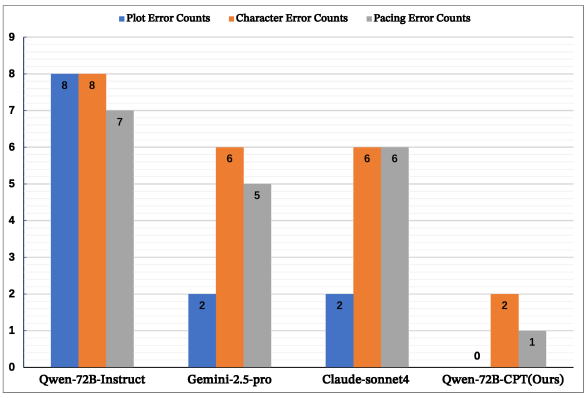

Figure 4: Error frequency comparison across models.

crucial—larger models significantly outperform smaller variants; (2) The two-stage DSR pipeline yields approximately 2.4 points improvement over end-to-end generation; (3) Both continual pre-training (CPT) and Chain-of-Thought (CoT) reasoning contribute to performance gains; (4) Our Hybrid data synthesis strategy outperforms the Reverse-only approach, achieving the lowest variance of 0.14. These findings collectively highlight the importance of appropriate model scale, domain-adaptive pre-training, and structured data synthesis in advancing LLMs for sophisticated creative applications.

5 Conclusions

In this paper, we focus on enabling large language models to generate high-quality screenplays from basic settings such as outlines and character profiles. Direct end-to-end generation is inadequate for this complex task: the generated screenplays may follow stylistic conventions, but often lack the deep structural integrity and storytelling substance required for professional use. Therefore, we propose the Dual-Stage Refinement (DSR) framework, which decomposes the single complex task into two distinct stages: creative narrative generation and strict format conversion. This decomposition allows models to focus on one specific skill at each stage, avoiding the poor performance caused by task entanglement. Implementing this framework requires paired training data for the narrative generation stage, which we obtain through an innovative hybrid data synthesis strategy. The efficacy of the combined DSR framework and data synthesis strategy was validated through extensive experiments. In blind evaluations conducted by professional screenwriters, screenplays generated by the DSR framework achieved a 75% win rate against strong baselines including Gemini-2.5-Pro, and reached 82.7% of human-level performance. These results demonstrate that a decomposed generation framework enabled by tailored hybrid data synthesis significantly outperforms approaches that rely solely on prompt engineering to guide large, general-purpose language models. As screenplay generation is representative of complex creative writing tasks, this decomposition-based approach is likely applicable to broader creative content generation scenarios.

Limitations

While our framework has proven effective, its current implementation has certain limitations. The data synthesis strategy relies on an initial corpus of high-quality human-written screenplays, which may limit its scalability to domains where such data is scarce. Future work will explore methods to reduce this dependency and enhance the scalability of the data synthesis process. Additionally, we plan to extend the DSR paradigm to other structured creative writing tasks, such as composing structured poetry or developing long-form fictional narratives, to further validate its generalizability across diverse creative domains.

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